Champfer Distance

Q4-1: Distance Transform (Manhattan)

The distance transform gives each pixel a value to its Manhattan distance to the nearest feature pixel. This is calculated using two passes over the image with programming. Feature pixels are set to 0. All other pixels start at infinity.

In the pass we go from top-left to bottom-right. For each pixel we update its value to be the smaller of its value and either (top neighbor + 1) or (left neighbor + 1).

Then in the pass we go from bottom-right to top-left. Each pixel updates to the smaller of its value and either (bottom neighbor + 1) or (right neighbor + 1).

Together these two passes compute the Manhattan distance for every pixel in O(N) time. This is much faster than checking every pixel, against every feature pixel. For the test image the resulting distance map is:

```
# Q4-1 answer block # - NO NEED TO CHANGE THIS PART, Run the cell to show the output.
image_features = np.array([
    [0, 0, 0, 1],
    [0, 0, 1, 0],
    [0, 1, 0, 0],
    [1, 0, 0, 0]
], dtype=np.uint8)


# Compute distance maps
dist_map_features = distance_transform_manhattan(image_features)

print("Distance Map for Image Features:")
print(dist_map_features)
```

```
Distance Map for Image Features:
[[3. 2. 1. 0.]
 [2. 1. 0. 1.]
 [1. 0. 1. 2.]
 [0. 1. 2. 3.]]
```

Q4-2: Chamfer Matching

The distance transform of the scene's edge image is calculated using the module from Q4-1. The template is then moved across every position (y, x) where it fits completely within the scene. At each position the chamfer distance is determined by taking the average of the distance-transform values that lie under the template's edge pixels. A low chamfer distance means the template edges are close to the edges, in the scene indicating a good match. The best match is the position where the chamfer distance's the smallest.

The best match was found at (y, x) = (139, 239) which corresponds to the right-angle triangle in the right of the scene. When the template is overlaid the red edges of the template. The diagonal, the top row and the right column. Align perfectly with the edges of that triangle. In the chamfer distance map, which is normalized to the range [0, 1] darker areas represent distances. The lowest value appears at the match location. The map has dimensions of 461 x 461 because the 40 x 40 template can only be placed in positions where it fits entirely within

the 500 x 500 scene.

